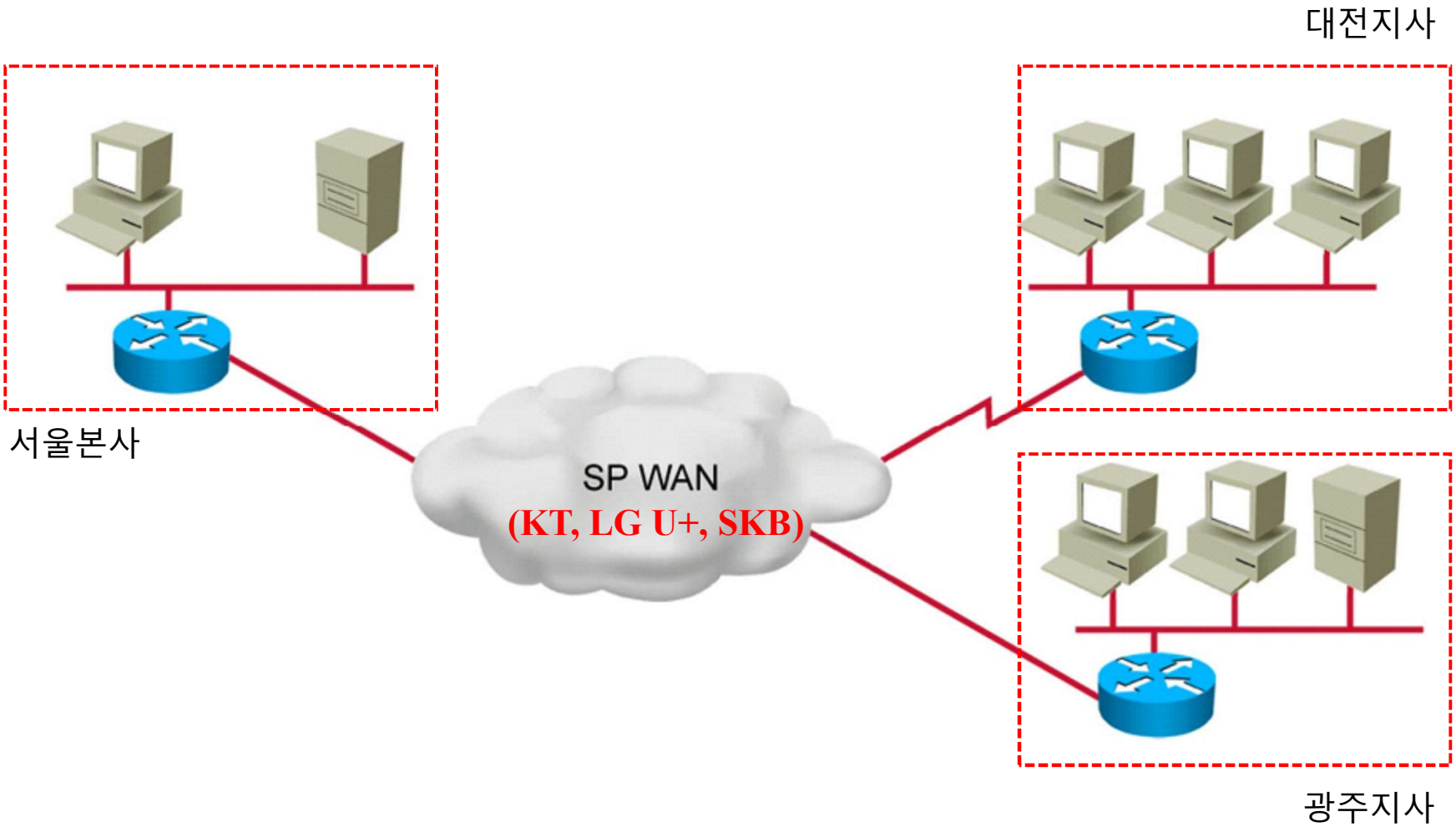
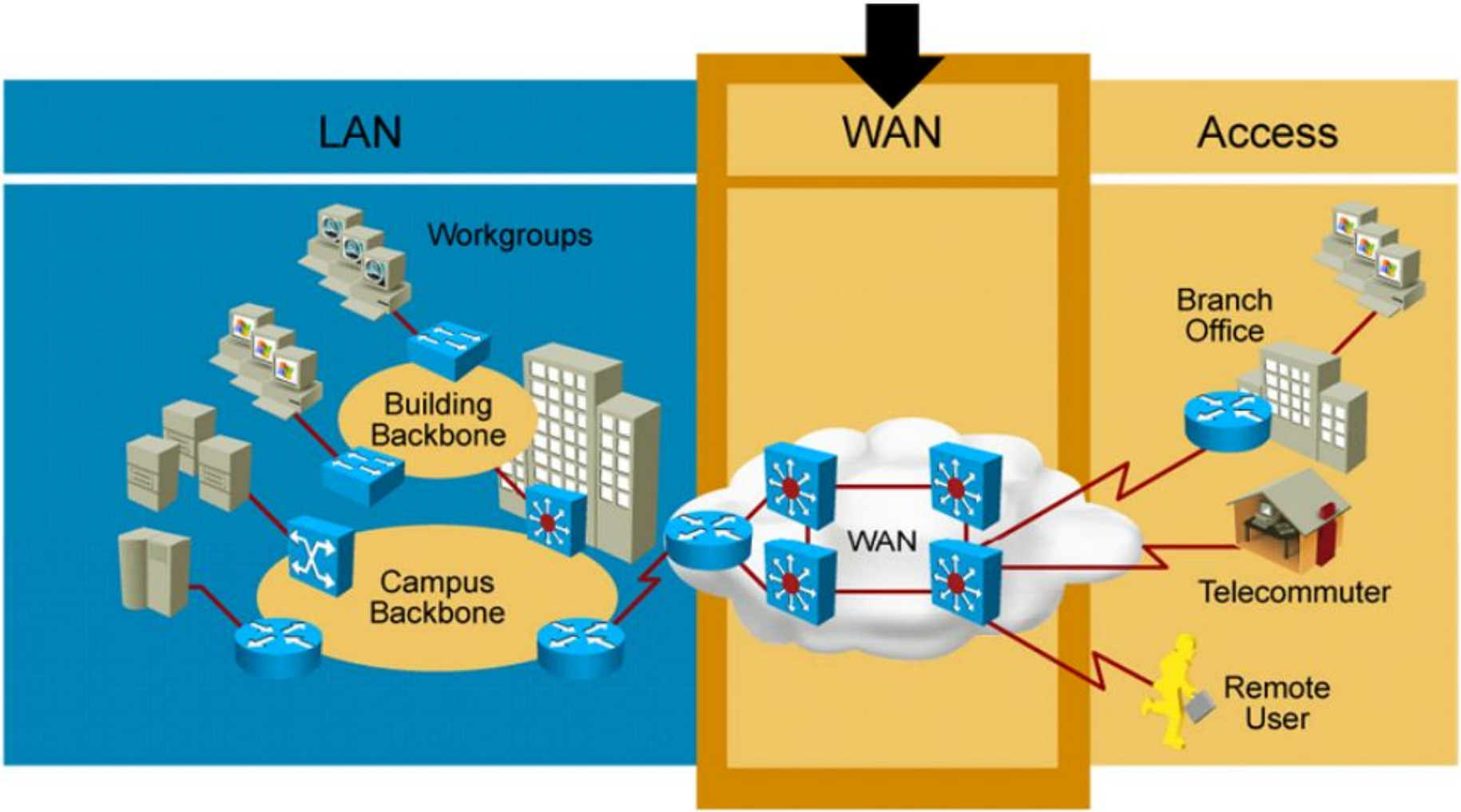
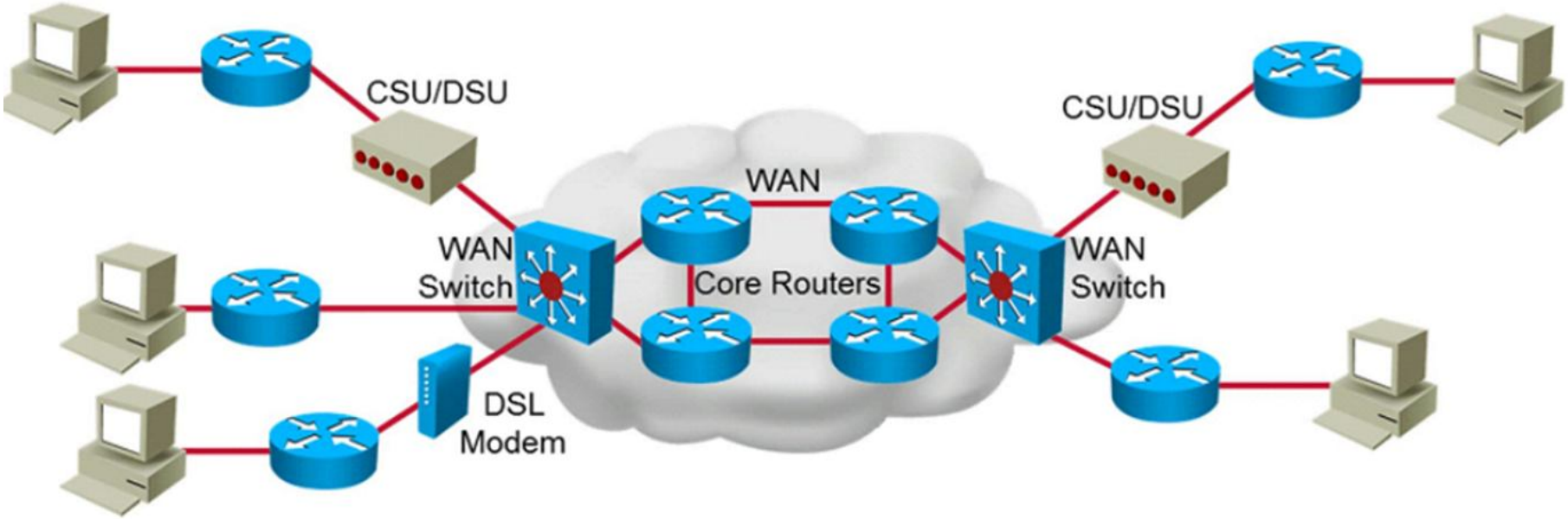
* WAN= A Collection of LANs



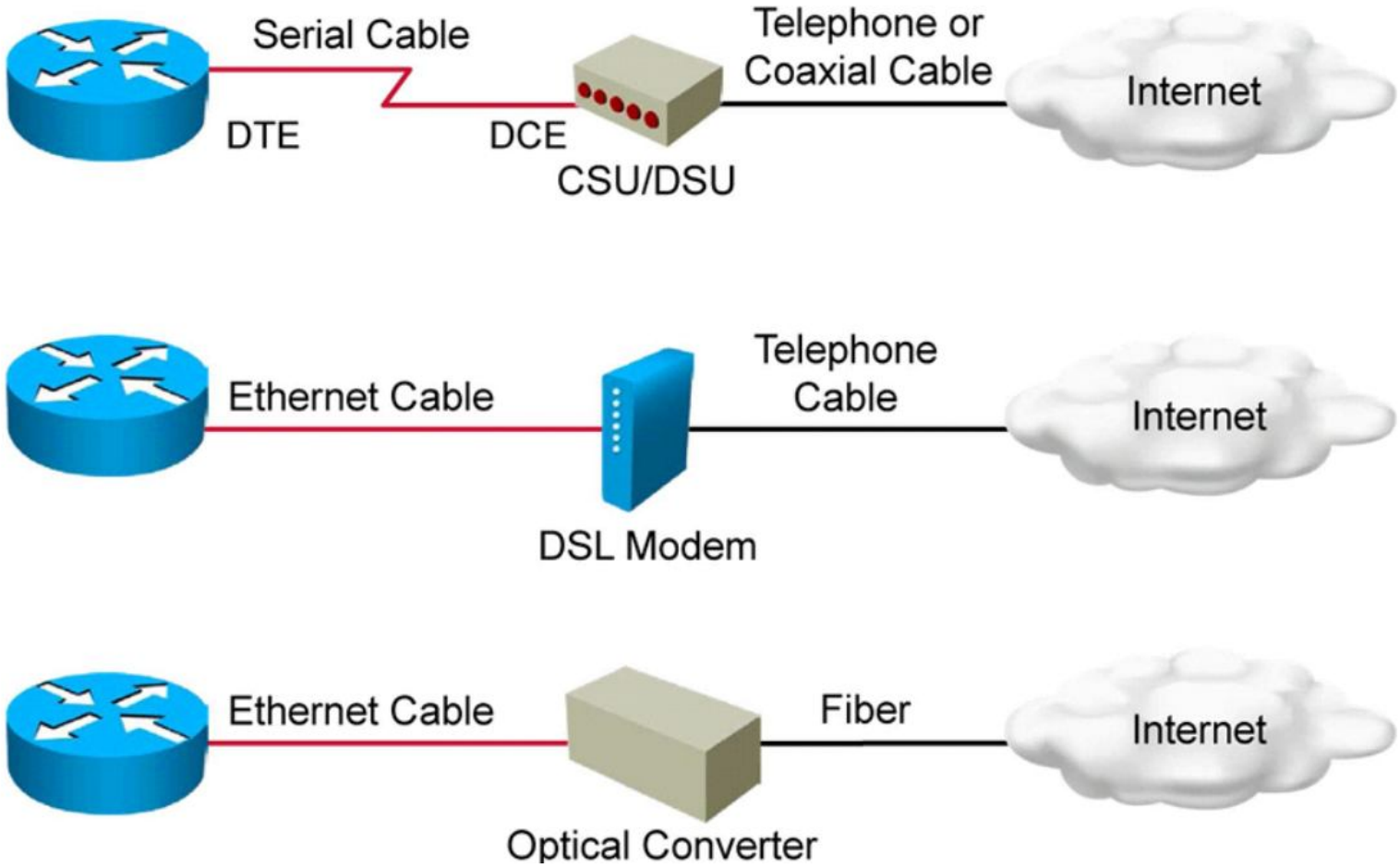

LAN
WAN
Access
Workgroups
Building
Backbone
Campus
Backbone
WAN
Branch
Office
Telecommuter
Remote
User

WAN Devices

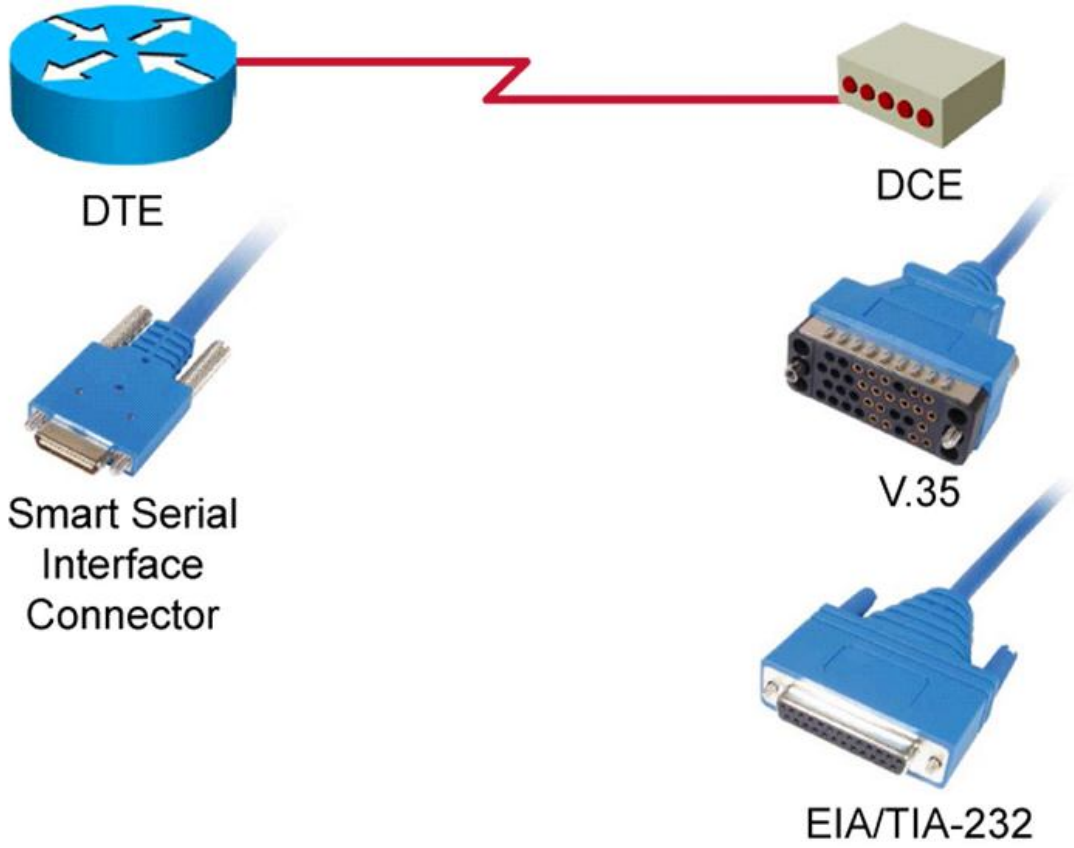
- Routers
- CSU/DSU
- WAN switches
- Core routers
- Modems



WAN Devices


Serial Cable
DTE
DCE
CSU/DSU
Telephone or
Coaxial Cable
Internet
Ethernet Cable
DSL Modem
Telephone
Cable
Internet
Ethernet Cable
Optical Converter
Fiber
Internet

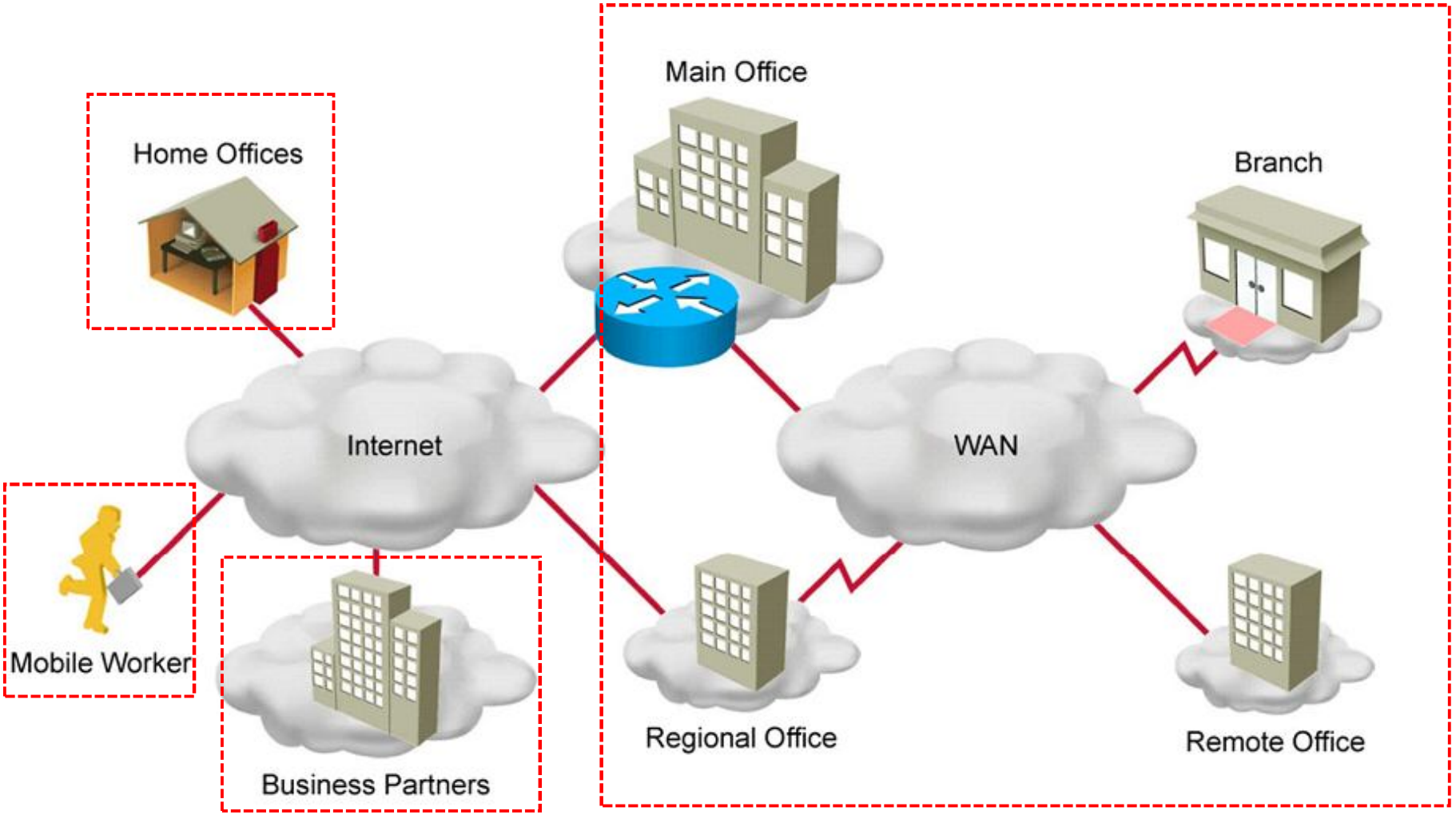
WAN Serial Cable



Serial Communication Links

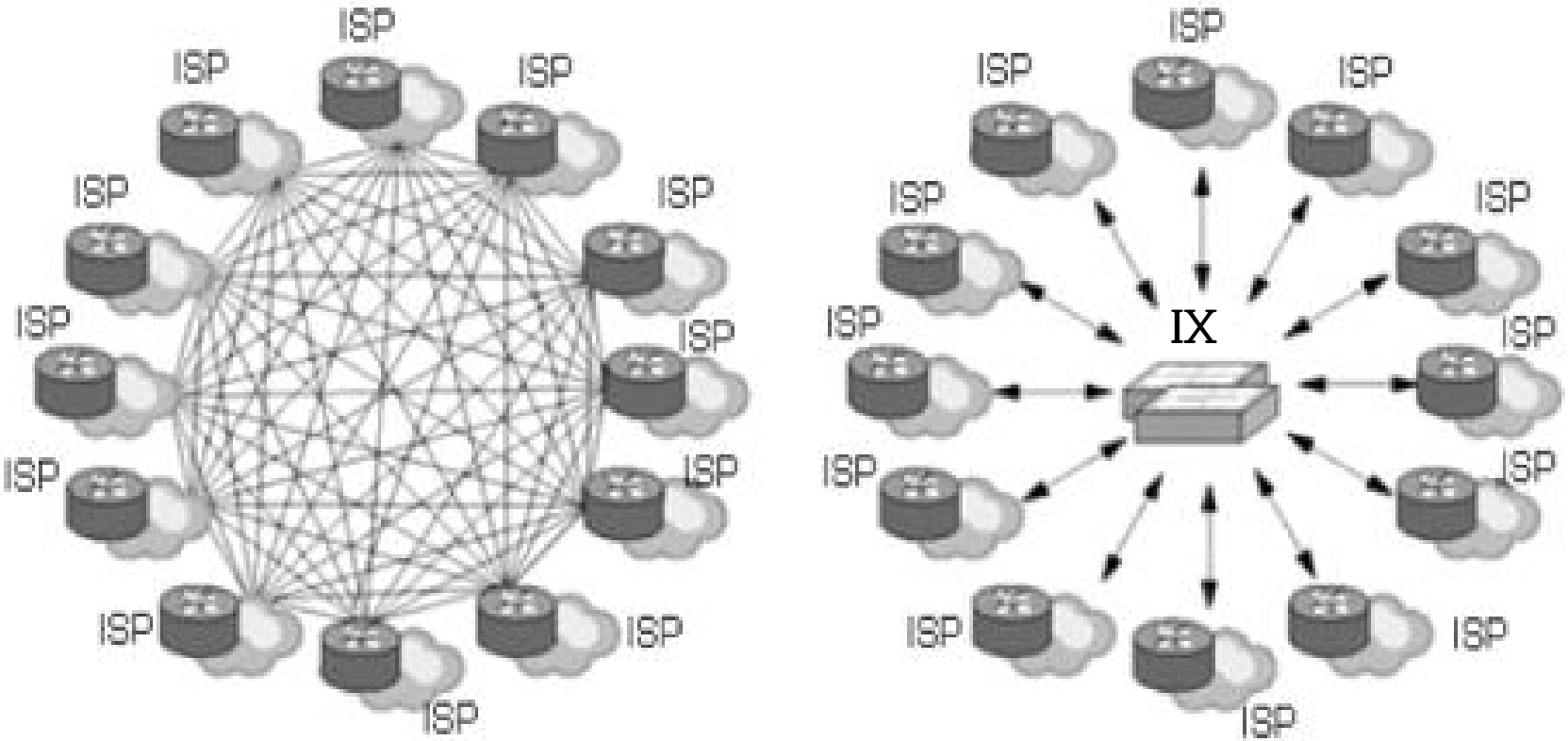
- Typical WAN speeds for the U.S.:
  - T1 = (1.544 Mb/s)
  - T2 = 4 T1 lines (6 Mb/s)
  - T3 = 28 T1 lines (45 Mb/s)
  - T4 = 168 T1 lines (275 Mb/s)
- Typical WAN speeds for Europe:
  - E1 = (2 Mb/s)
  - E2 = 128 E0 lines (8 Mb/s)
  - E3 = 16 E1 lines (34 Mb/s)
  - E4 = 64 E1 lines (140 Mb/s)

Internet & WAN



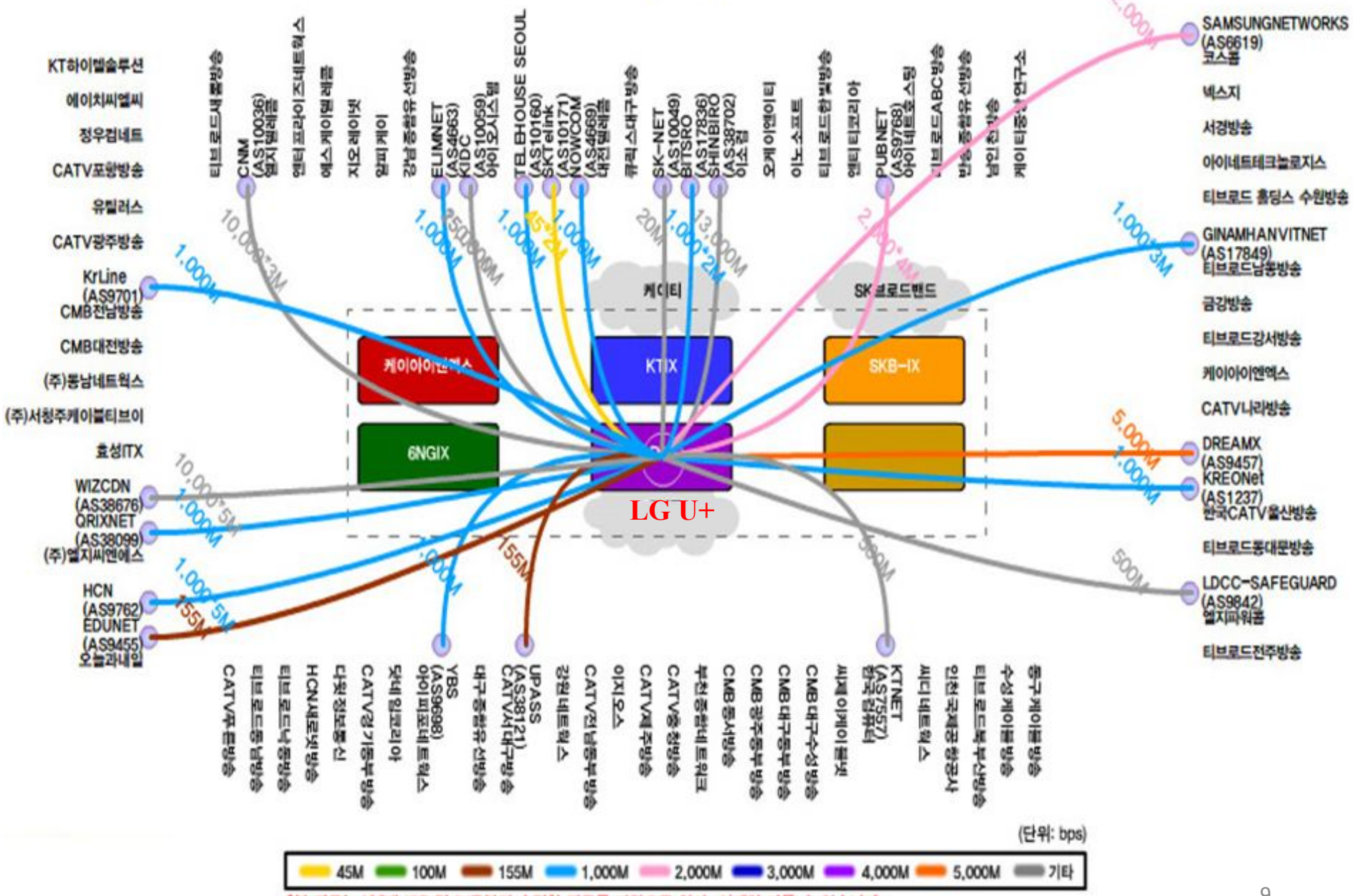
WAN= A Collection of LANs

IX(Internet eXchange)



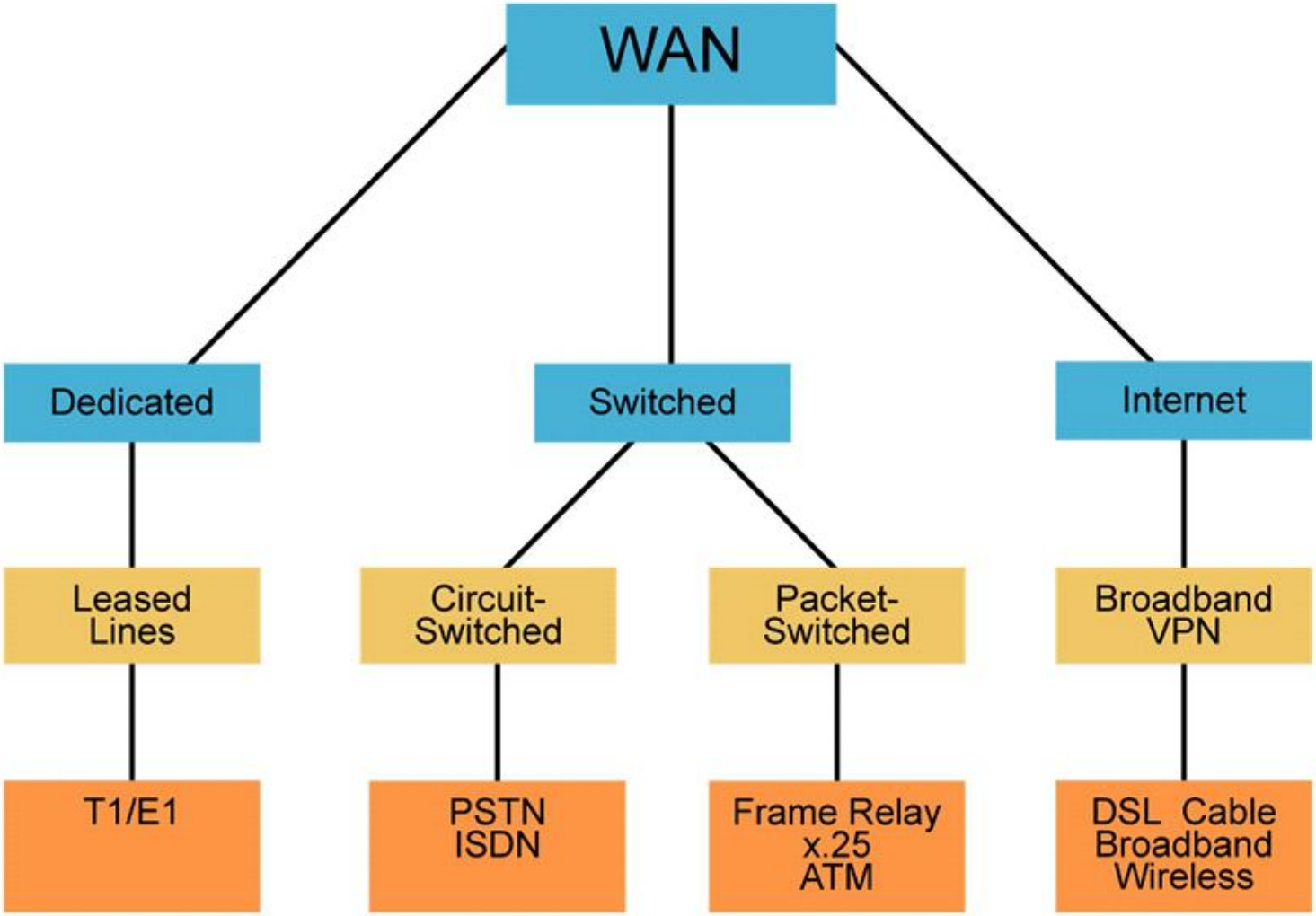
IX(Internet eXchange) -인터넷 트래픽을 상호 교환 시켜주는 역할 수행

Internet Connectivity Map Between ISP & IX

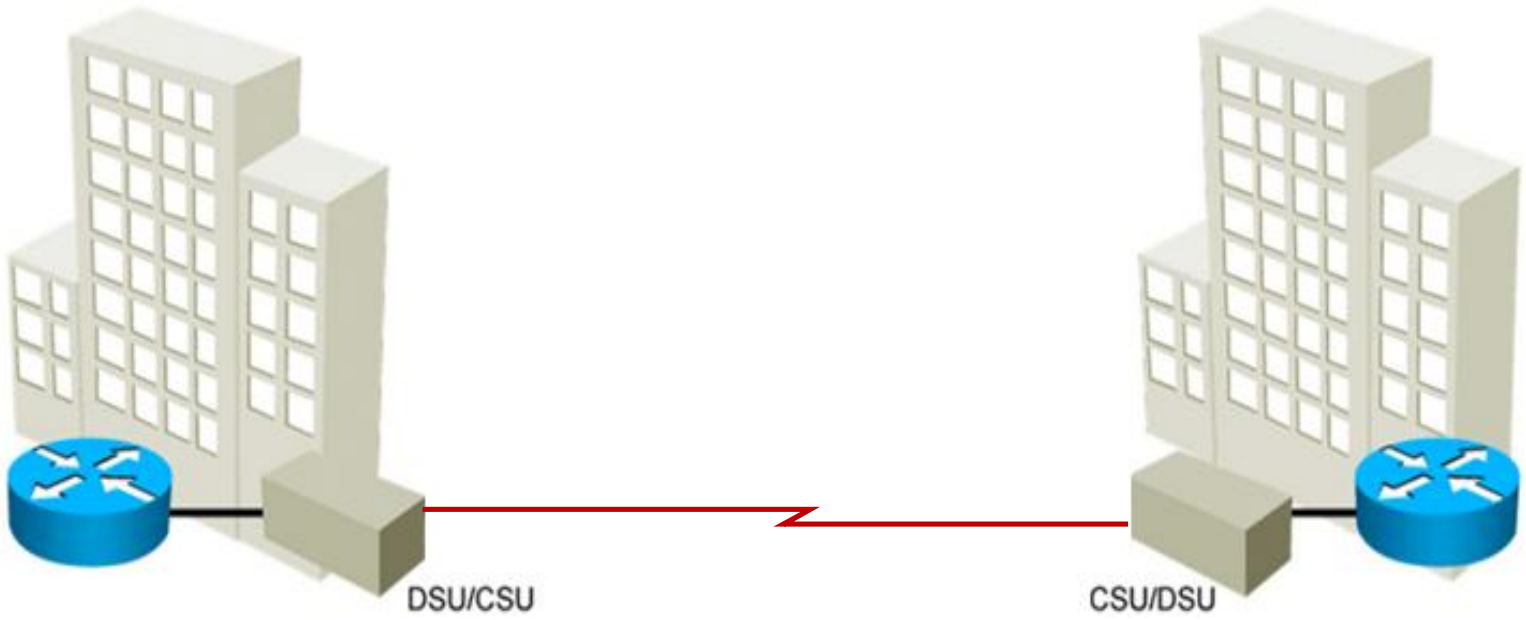


*본 자료는 10개 ISP 및 IX로부터 수집한 자료를 바탕으로 하며, 실제와 다를 수 있습니다.

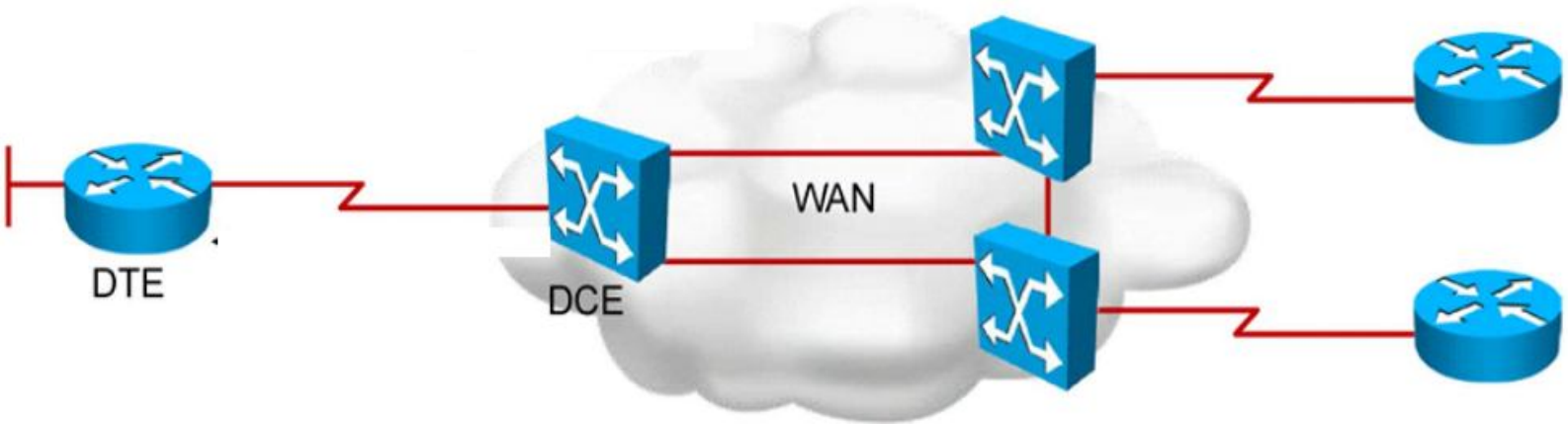
WAN(Wide Area Network) 구축 형태



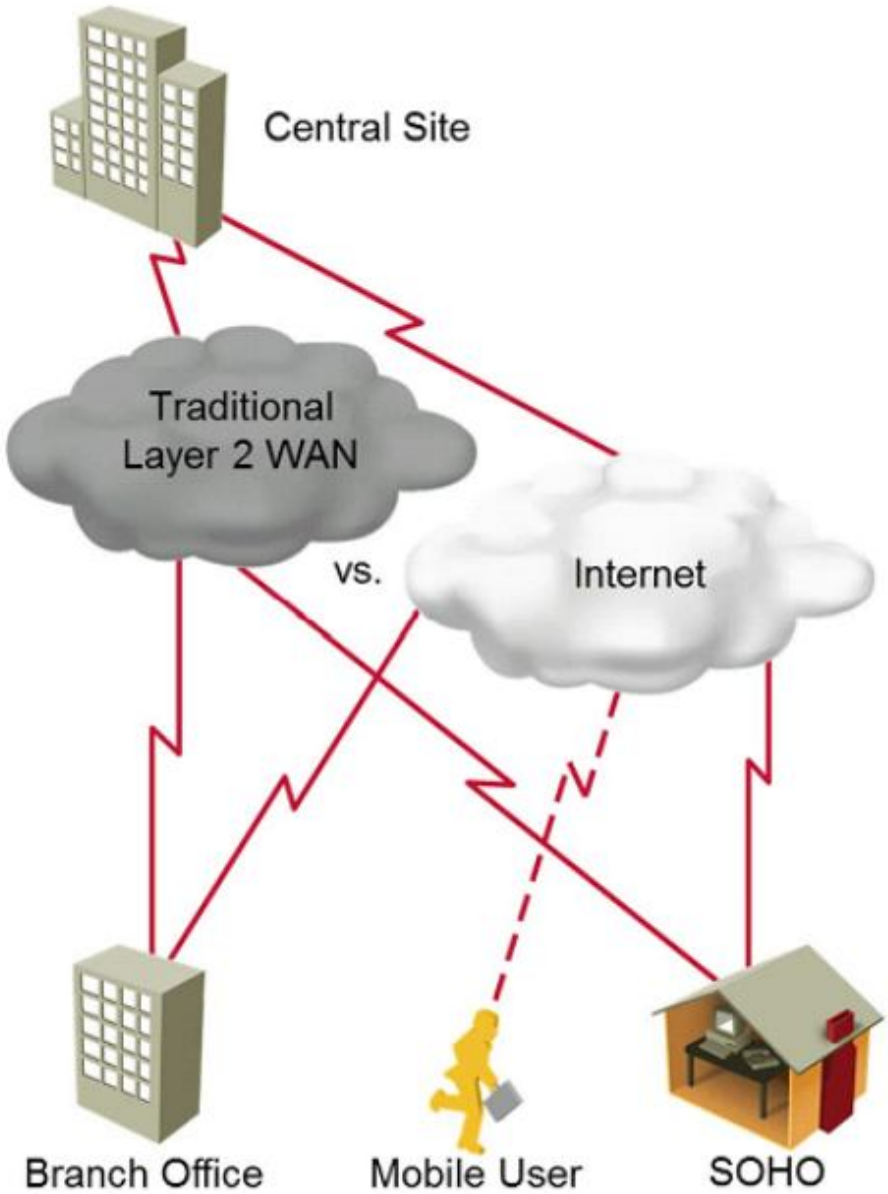
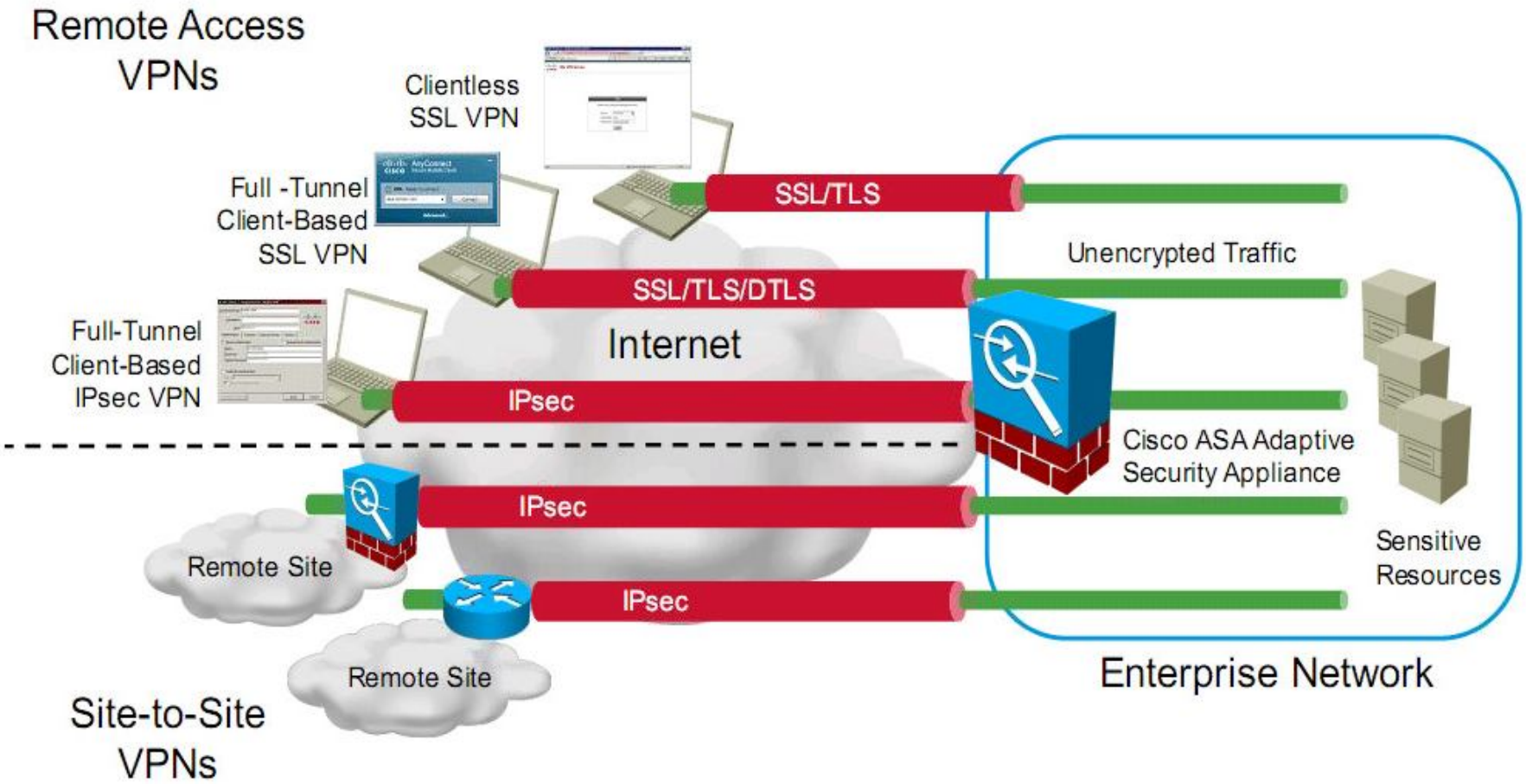
Dedicated(Leased lines)



Switched



Internet


Central Site
Traditional
Layer 2 WAN
vs.
Internet
Branch Office
Mobile User
SOHO

Virtual Private Networks (VPNs)